

Chelsea Dole

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[Github](#) ▪ [LinkedIn](#) ▪ [Website](#)

TECHNICAL SKILLS

- PostgreSQL (HA/DR, containerization, sharding, upgrades, performance)
- OLAP (Redshift, BigQuery)
- NoSQL (MongoDB, Redis)
- SQL Server
- Apache Kafka
- Kubernetes administration
- Python
- Golang
- Cloud (AWS, GCP)
- Terraform
- Docker
- GitOps (Helm, ArgoCD)
- Linux

WORK EXPERIENCE

Citadel, New York City

Database Engineer

July 2024–Present

Building database infrastructure, providing organization-level guidance about best practices, and optimizing performance for all teams at Citadel.

Technologies: PostgreSQL, Kubernetes, SQL Server, GCP, GitOps (Helm, ArgoCD), Python, Terraform, Linux

Brex, Remote (Seattle)

Staff Software Engineer (Tech Lead, Data Storage)

Aug 2023–June 2024

Senior Software Engineer

Feb 2022–Aug 2023

- Optimized read/write performance for PostgreSQL and Redis through query optimization, schema redesign, hash/range partitioning, configuration tuning, automation, and more.
- Contributed to planning & execution of widespread infrastructure cost saving initiatives resulting in 30% and 50% reduction in AWS RDS spend across production and staging respectively.
- Reduced a 7TB PostgreSQL database to 800GB through schema redesign and range partitioning with <100ms downtime, resulting in \$38K/year savings and 60% drop in read latency.
- Pitched, architected, and delivered “DB Metadata Service”, a DynamoDB/Golang-backed gRPC service enabling realtime software/hardware upgrades without impact on metrics or dependencies.

Technologies: PostgreSQL, DynamoDB, Redis, Golang, Terraform, PGBouncer, Kubernetes, AWS, Linux

Coffee Meets Bagel, Remote

Senior Data Engineer (Tech Lead)

May 2021–Jan 2022

Data Engineer III

April 2020–May 2021

Data Engineer II

June 2019–April 2020

- Architected and built distributed systems to support ML models and business-critical data warehouses
- Designed and fully implemented an event-based feature engineering system providing real-time features to machine learning models. Resulted in an 85% reduction in feature calculation time and estimated 75%+ reduction in engineering time needed to author features.
- Led project to shard a business-critical 3TB Postgres table with ~4B writes/day using Citus. Built multiple POC sharding options, performed stress testing, and estimated costs.
- Planned & implemented a company-wide migration from RabbitMQ/Kinesis to Apache Kafka.

Technologies: PostgreSQL (PGBouncer, Patroni, Citus), Python, Kafka, Elasticsearch, AWS, Django, RabbitMQ, Kubernetes, Redis, Docker, Linux

Zonar Systems, Seattle

Software Development Engineer I

Dec 2017–June 2019

Built backend applications to support a data analysis web platform, including components of a Python data analysis API, a Golang event logging API, and a Dataflow pipeline.

Technologies: Python, Golang, JavaScript/Node.js, Kubernetes, PostgreSQL, GCP, Pandas/NumPy

SPEAKING

Postgres Platform “Best Practices” for the Modern DBA ([slides](#))

50-minute talk about Postgres platform-building. Given at: PGConf NYC 2024, PGConf EU 2024.

Breaking Up with Massive Tables via Partitioning ([slides](#), [video](#))

50-minute talk about how to partition live, production-critical PostgreSQL tables while minimizing downtime. Given at: PGConf NYC 2023, PGConf EU 2023, Seattle & NYC PUGs (2025, 2026).

Postgres at Scale: Vacuum, MVCC, & Table Bloat ([slides](#), [video](#))

30-50 minute talk about understanding, quantifying, and mitigating PostgreSQL “table bloat”. Given at: SoCal Linux Expo 2023, PGDay Paris 2023, Nordic PGDay 2023, PASS Data Summit 2023, etc.

Understanding Python's Memory Management ([slides](#), [video](#))

Short introduction to Python memory management. Given at: PyLadies Int'l Women's Day 2021.

Dreaming of Databases: 5 Questions for Data Layer Decisions ([slides](#))

25-minute talk about database vocabulary and architecture. Given at: ThaiPy Bangkok 2020.

PROJECTS // OPEN SOURCE

[PostgresConf NYC](#), Organizer

2024-Present

One of four primary organizers of the PostgresUS Foundation's largest annual conference. Responsible for talk selection, program/schedule management, volunteer coordination, and more.

[PostgresUS Foundation](#), Chair (Diversity Committee)

2023-2025

Responsible for allocation and management \$20K annual funding with the goal of supporting PostgreSQL community members from underrepresented backgrounds.

EDUCATION

Code Fellows, Seattle, WA

2017

Certificate: Backend Software Development

University of California, Berkeley

2013–2017

Degree: Political Economy (B.A.)

GPA: 3.40